

Charles Forsyth

Director of Research Computing
University of California, Riverside

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Summary

Research computing leader in high-performance computing (HPC) and artificial intelligence/machine learning (AI/ML) with more than 25 years of experience in system administration, software development and research IT. As Director of Research Computing at the University of California, Riverside, I lead the design, implementation and optimization of the HPC systems and applications that drive research across UCR's disciplines, including AI and ML, along with the training programs and research support that build the skills of our faculty and students.

My career spans R1 universities, a Fortune 500 research campus, systemwide University of California leadership and United States Navy electronics. My expertise covers HPC system administration, AI/ML integration, cloud research computing, software development and documentation, license management and systems design.

Professional Experience

● Director of Research Computing

Mar 2023 – Present

University of California, Riverside · Riverside, CA

- **Technological Leadership and Innovation:** Led the integration and implementation of cutting-edge technologies (HPC clusters, GPUs, FPGAs, AI, ML, containerization) across UCR's scientific disciplines, significantly enhancing research computing capabilities.
- **Ursa Major:** Lead Ursa Major, UCR's research computing partnership with Google Cloud, giving researchers high-performance workstations, auto-scaling HPC clusters, research AI and machine learning services and secure research storage.
- **Campus-Wide Research Integration:** Unified six schools and nine centers to collaboratively support 800+ research labs, fostering a comprehensive and integrated research computing environment.
- **Grant Writing:** Secured multiple grants through active engagement in grant writing, funding UCR's research computing infrastructure and initiatives.
- **ACCESS Program Advocacy:** Championed UCR's role in the NSF ACCESS program, promoting advanced computing in research.
- **Strategic Partnerships and Representation:** Represented UCR in key research computing consortia and platforms, including the Campus Research Computing Consortium (CaRCC) and the Pacific Research Platform.
- **UC Research IT Committee:** Voting member aligning research IT and HPC strategies across the UC campuses and with the UC Office of the President.

- **Chair, UCR Research IT Leaders Committee:** Led strategic planning and implementation of key research computing projects aligned with UCR's academic and research goals.
- **Strategy and Infrastructure Oversight:** Manage the development and continual enhancement of UCR's research computing strategy and infrastructure as research needs evolve.

● **Associate Director of Research Computing**

Aug 2019 – Mar 2023

University of California, Riverside · Riverside, CA

- **Strategic Thought Leadership:** Advanced research computing within Information Technology Solutions (ITS), developing grant strategy and managing budgets for research technology initiatives.
- **Systems Architecture:** Lead systems architect for an environment built around research computation, advanced statistical computing and network interoperability with the campus Science DMZ, ensuring high standards of operation and security.
- **Operational Leadership:** Key decision-maker in technology discussions, serving as a proxy for the Chief Information Officer or Chief Technology Officer in academic senate technology discussions.
- **Research and Development Oversight:** Directed a team of professionals in high-level HPC, data and cyberinfrastructure research and technology projects.
- **Team Supervision and Policy Management:** Managed and mentored technical staff across fiscal, HR and resource allocation policy.
- **Collaboration and Integration:** Coordinated research technology efforts within ITS with BCOE, HPCC and library partners, aligned with UCR's academic and research objectives.

● **Chairman, UC Research IT Committee**

Aug 2021 – Aug 2022

University of California, Office of the President

- Led the UC Research IT Committee, liaising directly with the IT Leadership Council (ITLC) and regularly with the UCOP Chief Information Officer.
- Brought together research IT professionals from every UC campus into a committee focused on strengthening UC's research IT landscape.
- Drove initiatives aligned with the UC Office of the President's vision, elevating research IT infrastructure and capabilities across the UC system.
- Built a strong community among the groups supporting research IT, promoting shared knowledge and best practices across campuses.
- Advocated for the needs of University of California researchers and met the committee's annual goals.

● **Lead Linux & HPC Systems Administrator / Consultant**

Jun 2017 – Aug 2019

University of California, Riverside · Riverside, CA

- Architected, installed, administered and supported the campus HPC systems.
- Trained researchers and students through classes, seminars and hands-on solutions.
- XSEDE Campus Champion, consulting and enabling researchers and students to access NSF-backed supercomputing centers and national laboratories.
- UCR representative to the University of California Research IT Committee, aligning Research IT and HPC goals across the UC campuses and the UC Office of the President.
- Led the implementation of functional on-demand HPC clusters in AWS.
- UCR representative to the Campus Research Computing Consortium (CaRCC).

Lead Linux & HPC Systems Administrator / Research Consultant

Jan 2012 – Jun 2017

Corning Incorporated · Corning, NY

- Lead HPC administrator architecting, installing, administering and supporting large-scale HPC infrastructure for R&D operations: Linux clusters with thousands of cores and Lustre, SAN and NAS storage.

PROVISIONING

iLO, Cobbler, iPXE, Kickstart

MONITORING

Nagios, ELK stack

STORAGE

HP MSA SAN, Lustre, Isilon NAS, DDN Exascaler, GPFS / IBM Spectrum Scale

NETWORKING

InfiniBand, 10 GbE and 1 GbE, 16 Gb Fibre Channel

VIRTUALIZATION

VMware, Citrix XenServer, oVirt KVM

OPERATING SYSTEMS

CentOS Linux 4 to 7, Windows XP to Server 2012 R2

PROGRAMMING

C++, C#, Python, Bash, PowerShell

REMOTE DESKTOP

VNC, RDP, X11, NICE DCV

SOURCE CONTROL AND DATABASES

Git, MySQL, MS SQL

SERVER SOFTWARE AND CLOUD

SSH, cgroups, Apache, OpenStack, AWS

SCIENTIFIC SOFTWARE

Ansys, COMSOL, Abaqus; Intel MPI, OpenMPI; MKL, FFTW

LICENSE MANAGEMENT

FlexLM, RLM, LM-X, MathLM for 60+ network-licensed packages

FACILITIES

150 kVA+ APC UPS, large Liebert cooling units

CONTRACTS

400+ server, storage and networking support contracts

LEADERSHIP

Internal and external project management, customer training and documentation

Research System Administrator / Engineer / Owner

Jun 2003 – Jan 2012

CnL Networks · Whitney Point, NY

- Founded and operated a systems administration support and outsourcing business delivering IT solutions to a diverse client base.
- Managed every part of the business, including billing, accounting, infrastructure development and support systems.
- Provided system design, training and customer support, including research computing support, web design, network setup and procurement consulting.

- Deployed and maintained Linux and Windows server environments and built e-mail and web hosting solutions for clients.

● **Technical Account Manager**

Apr 2008 – Aug 2009

STCR · Endicott, NY

● **System Administrator / Engineer**

Mar 1998 – Jun 2003

Hancock Telephone · Hancock, NY

- Provided internal support and customer service for internet connectivity, hosting and outsourced systems.
- Secured and managed network infrastructure, logistics and operations.
- Designed and managed Windows and Linux server environments (web, email, NAS, DNS, RADIUS and file servers) plus firewalls, IDS, asset management and packet inspection.

● **Hardware Test Engineer**

Aug 1997 – Mar 1998

IBM · Endicott, NY

- Tested and troubleshoot hardware components and systems with a focus on root cause analysis.
- Analyzed and documented performance trends to improve production processes.

● **Technical Support**

Mar 1997 – Aug 1997

America Online · Jacksonville, FL

- Delivered frontline technical support for dial-up and DSL connections.
- Led internal training sessions to build team capabilities.

● **Advanced Electronics Technician**

Dec 1994 – Mar 1997

United States Navy · USS Lake Champlain (CG-57), Norfolk, VA

- Troubleshoot and maintained radar and communications systems, including IFF systems, from system to component level.
- Trained operators on system functionality and operations.
- Managed hardware records, maintenance and testing for IFF systems.

Education

● **Bachelor of Science, Computer / Information Technology Administration and Management**

2013 – 2015

Purdue Global (formerly Kaplan University)

- GPA 3.89
- Iota Sigma Tau Honor Society, Golden Key International Honor Society, Alpha Beta Kappa National Honor Society

● **Networking and Technical Support Specialist**

2010 – 2011

Ridley-Lowell Business & Technical Institute · Binghamton, NY

- GPA 4.0; Dean's List; top GPA of the class

- Training for supporting end users and computer operations: hardware and peripherals, networking, operating systems and software applications
- Qualified certifications equivalent to CompTIA A+, Network+, Security+ and Linux+, and Microsoft Certified Systems Engineer (MCSE)

Advanced Electronics Technology / Technician

1994 – 1997

United States Navy

AC and DC Circuits

Basic and Advanced Electronics Laboratories

Digital Microprocessor Fundamentals

Electronic Communications

Electronic Systems Troubleshooting

Radio / Radar Fundamentals

Solid State Electronics

Analog and Digital Circuit Analysis

Communications Circuit Analysis

Digital Circuit Logic

Teaching and Documentation

Training decks, terminal recordings and a full HPC field manual (Linux, hardware, provisioning, software, code, cloud clusters and Slurm) are published in the [HPC Field Manual](#), with talks collected under [Talks and Presentations](#).